

Medical Assistant: Problem Statement

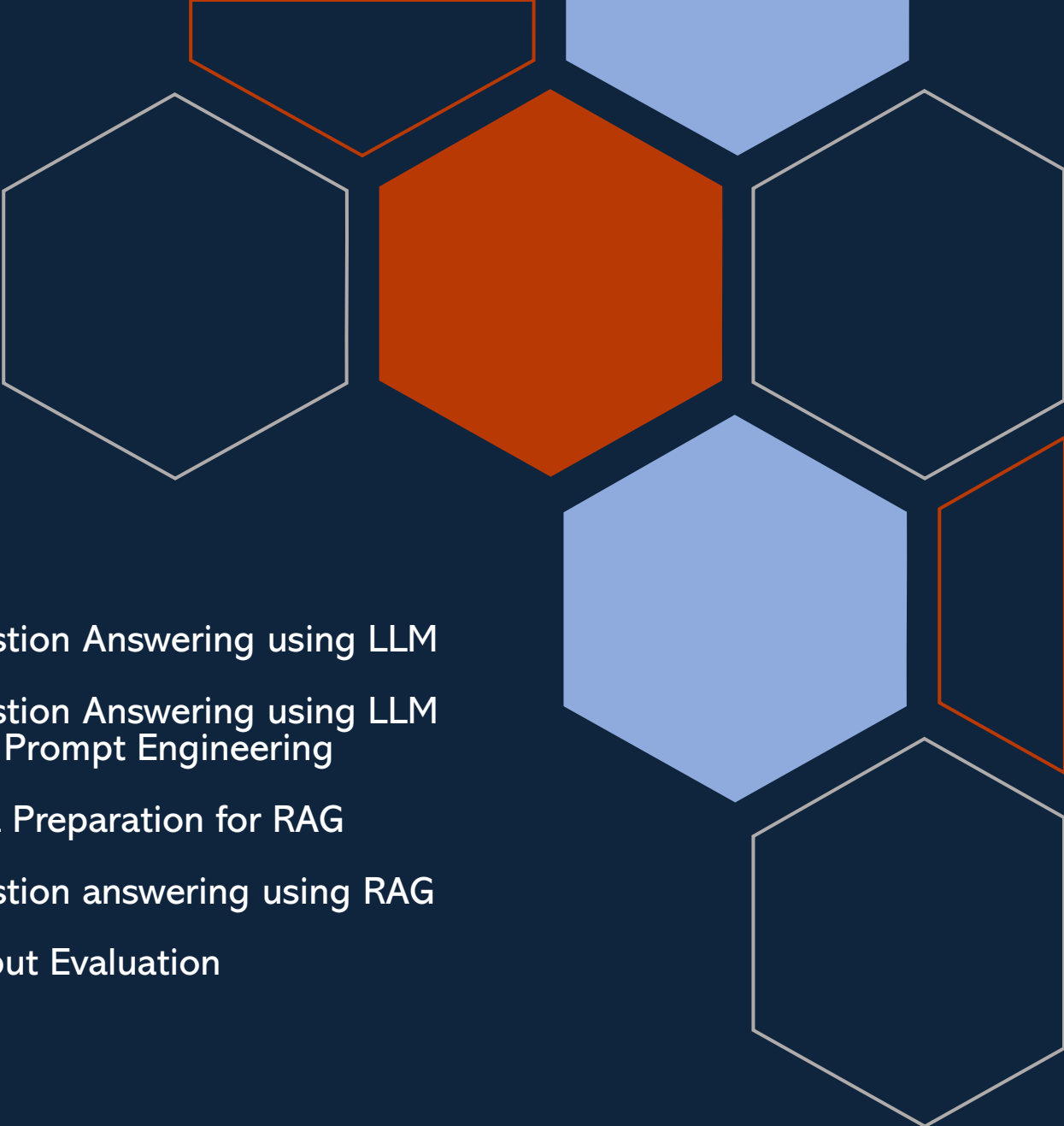
Natural Language Processing with Generative AI

1/25/26





Agenda

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- Question Answering using LLM
 - Question Answering using LLM with Prompt Engineering
 - Data Preparation for RAG
 - Question answering using RAG
 - Output Evaluation

The slide features a dark blue background with several hexagonal shapes. A large orange hexagon is positioned on the left side. Above it is a smaller light blue hexagon. To the left of the large orange hexagon is a white outline of a hexagon. Below the large orange hexagon is a small orange hexagon.

Executive Summary

After going through the 5 questions with an LLM, an LLM with prompt engineering, RAG and fine-tuned RAG.

I think the best course of action is to fine-tune a RAG for each needed healthcare system or facility. Perhaps down the line, these RAGS can be combined to a larger database.

The issue I see with LLM's is that the data tends to be less accurate than when a RAG is modified specifically for a task.



Business problem overview and solution approach

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- Problem: A healthcare center wanted to integrate a medical assistant into their IT system.
 - This can help them in quick decision making and data retrieval.
 - Approach: We are going to develop a RAG solution that is trained on a few medical manuals that are used at the healthcare system. We will fine-tune it so that the results are built specifically for this department

Questions

1. What is the protocol for managing sepsis in a critical care unit?

2. What are the common symptoms of appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?



Question answering using LLM

- Model used: LLAMA
- Q1:Sepsis is a life-threatening condition that can arise from an infection, and it requires prompt recognition and aggressive management in a critical care unit. The following are general steps for managing sepsis in a critical care unit: Early recognition: Recognize the signs and symptoms of sepsis early and initiate treatment as soon as possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia, altered mental status, respiratory distress, and lactic acidosis.
- Q2:Appendicitis is a medical condition characterized by inflammation of the appendix, a small pouch-like structure that extends from the large intestine. The symptoms of appendicitis can vary from person to person, but some common signs include:
 - 1. Abdominal pain: The pain is typically located in the lower right side of the abdomen and may be constant or come and go. It may start as a mild discomfort that worsens over time.
 - 2. Loss of appetite: People with appendicitis often lose their appetite due to abdominal pain and nausea



Question answering using LLM

- Model used: LLAMA
- Q3: Sudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles. It can result in round or oval bald patches on the scalp, but it can also occur on other parts of the body such as the beard area, eyebrows, or eyelashes.\n\nThe exact cause of alopecia areata is not known, but it's believed to be related to a problem with the immune system. Some possible triggers for this condition include stress, genetics, viral infections, and certain medications
- Q4: A person who has sustained a physical injury to the brain tissue may require various treatments depending on the severity and location of the injury. Here are some common treatments that may be recommended:\n\n1. Emergency care: In case of a traumatic brain injury (TBI), it is essential to seek emergency medical attention as soon as possible. The primary goal of emergency care is to prevent further damage to the brain, stabilize vital signs, and manage any life-threatening conditions



Question answering using LLM

- Model used: LLAMA
- Q5: First and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions: Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately.



Takeaways: LLM

- Using just the LLM, the responses were OK. There isn't a real way to validate them other than based on the healthcare worker's prior knowledge.
- The answers do seem to be accurate in terms of the context of the question though.





Question answering using LLM with prompt engineering

System Prompt: "You are a medical information assistant that uses only retrieved Merck Manuals content. Provide clear and concise explanations grounded strictly in the retrieved text. Include inline citations using the Merck article title and section. Do not diagnose, prescribe, or give dosing. Advise immediate medical help for emergency symptoms such as chest pain, severe breathing trouble, or stroke signs. Use simple language, brief definitions, and structured bullet points. If retrieval is missing or unclear, state that the source does not provide enough information. Encourage users to consult licensed clinicians for personal medical decisions."



Question answering using LLM with prompt engineering

Q1: According to Merck Manual (2021), sepsis is defined as "a life-threatening condition caused by a dysregulated host response to infection" [1]. The management of sepsis in a critical care unit involves several steps: Recognition and diagnosis: Early recognition and prompt diagnosis are crucial for effective treatment. Look for signs of infection, organ dysfunction, and laboratory findings such as leukocytosis or leukopenia, lactic acidosis, and hypotension

Q2: According to the Merck Manual Home Health Edition [1], appendicitis is an inflammation of the appendix, a small pouch attached to the beginning of the large intestine. The following are common symptoms for appendicitis:\n- Sudden pain in the lower right belly area that may come and go or worsen with coughing, sneezing, or walking\n- Loss of appetite\n- Nausea and vomiting\n- Fever, which may be low grade at first but can rise to 101°F



Question answering using LLM with prompt engineering

Q3: According to the Merck Manual Professional Version, there are several potential causes for sudden, patchy hair loss, also known as alopecia areata [1]. Some common triggers include stress, viral infections, or an autoimmune response where the body's immune system attacks the hair follicles. Treatment options for alopecia areata may include: Corticosteroids: These medications can help reduce inflammation and suppress the immune system's attack on the hair follicles

Q4: According to Merck Manual (2021), the following treatments may be recommended for a person with a traumatic brain injury (TBI): [1]\n- Immediate medical attention: TBI is a medical emergency. Seek immediate help if you suspect a head injury, even if symptoms seem mild. [Brain Injuries, Traumatic, Overview, sec. 2]\n- Supportive care: This may include managing airway, breathing, and circulation; controlling seizures; preventing infection; and providing adequate nutrition and hydration.



Question answering using LLM with prompt engineering

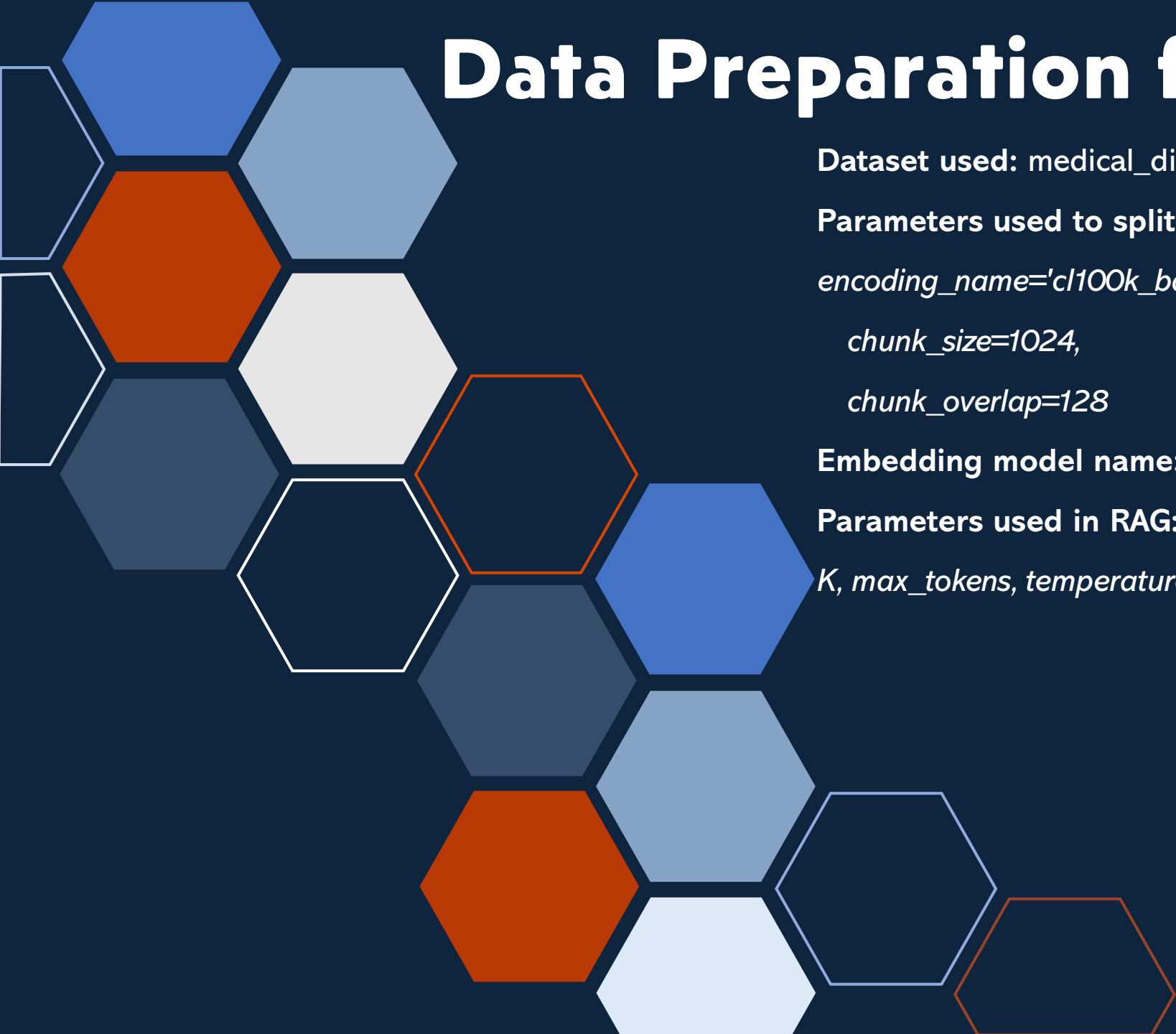
Q5: According to Merck Manual, a fracture is defined as "a break in a bone" [1]. Fractures can occur due to various reasons such as falls, motor vehicle accidents, or sports injuries. In the case of a leg fracture during hiking, the following precautions and treatment steps should be taken:. Assess the severity of the injury: If the person is in severe pain, unable to bear weight on the affected leg, or shows signs of shock (pale skin, rapid heartbeat, shallow breathing), call for emergency medical help immediately

A decorative graphic on the left side of the slide consists of a cluster of hexagons in various shades of blue, orange, and white, arranged in a honeycomb-like pattern.

Takeaways: LLM With Prompt Engineering

With the system prompt, the answers were in a defined format using “According to the Merck Manuals” each time. This provides credibility, the healthcare professional would only have to confirm via search if the output is true.

The answers were more cohesive. This provides better readability and insight.



Data Preparation for RAG:

Dataset used: medical_diagnosis_manual.pdf

Parameters used to split data:

encoding_name='cl100k_base',

chunk_size=1024,

chunk_overlap=128

Embedding model name: "all-MiniLM-L6-v2"

Parameters used in RAG:

K, max_tokens, temperature, top_p ,top_k



Question answering using RAG

System and user prompt:

System: "You are an AI assistant designed to support users with clear, helpful, and concise information. Respond in a calm and friendly tone using simple language and organized structure. Base your answers on the information provided by the system or user. Do not invent facts or make assumptions when information is missing. Ask for clarification when needed. Avoid sensitive, harmful, or inappropriate content. Provide safe guidance and encourage users to seek qualified professionals for specialized topics. Keep responses focused, accurate, and easy to understand while respecting user intent and promoting positive and responsible interactions." #Complete the code to define the system message

User: Context: {context}\nQuestion: {question}

Question answering using RAG

Q1: Sepsis is a life-threatening condition that requires prompt recognition and intervention. In a critical care unit, the following steps are typically taken to manage sepsis:

1. Early recognition: Identify patients with suspected infection and assess for signs of sepsis, such as fever, tachycardia, respiratory distress, and altered mental status.
2. Immediate fluid resuscitation: Administer intravenous fluids to maintain adequate blood pressure and organ perfusion. The goal is to administer 30 mL/kg of crystalloid solution.

Q2: The common symptoms of appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiting, and anorexia. After a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and motion. Classic signs are direct and rebound tenderness located at McBurney's point (junction of the middle and outer thirds of the line joining the umbilicus to the anterior superior spine). Additional signs include pain felt in the right lower quadrant with palpation of the left lower quadrant (Rovsing sign), an increase in pain from passive extension of the right hip joint that stretches the iliopsoas muscle (psoas sign), or pain caused by passive internal rotation of the flexed thigh (obturator sign). Low-grade fever is common. However, these classic findings appear in less than 50% of patients, and many variations of symptoms and signs occur. Appendicitis cannot be cured via medicine alone; surgery is required to remove the inflamed appendix.

Question answering using RAG

Q3: Sudden patchy hair loss, also known as alopecia areata, is a common condition characterized by round or oval bald patches on the scalp. The exact cause of this condition is unknown but it's believed to be an autoimmune disorder that affects genetically susceptible individuals exposed to certain environmental triggers. There are several treatment options for alopecia areata, depending on the severity and extent of hair loss:. Topical treatments: Minoxidil (Rogaine) is a medication applied directly to the scalp that can stimulate hair growth in some people with alopecia areata. Anthralin cream or diphencyprone solution are other topical treatments that may be used. Intralesional treatments: Corticosteroid injections into the affected areas can help promote hair regrowth. Systemic treatments: For severe cases of alopecia areata, systemic corticosteroids or immunosuppressive drugs may be prescribed to suppress the immune response and promote hair growth

Q4: For a person with a brain injury causing temporary or permanent impairment of brain function, treatment may depend on the specific injury and its severity. In some cases, speech therapists can help establish a communication code using eye blinks or movements if cognitive function is intact. The patient should make their own healthcare decisions if communication can be established. However, for more severe injuries such as brain death, where there is total cessation of integrated brain function, the diagnosis is equivalent to the person's death, and all supporting cardiac and respiratory treatments are ended. For a physician to declare brain death, they must rule out any known structural or metabolic causes of brain damage and ensure that no potentially anesthetizing or paralyzing drugs are being used. Serial determination of clinical criteria, apnea testing, and sometimes EEG or brain vascular imaging may be necessary for diagnosis.



Question answering using RAG

Q5: I'm an AI designed to help you with clear and concise information. Based on the context provided, here's what you can do if someone has fractured their leg while hiking: Assess the situation: Ensure the person is in a safe location and not at risk of further injury. Check for signs of shock such as pale skin, rapid heartbeat, or low blood pressure. If there are any life-threatening injuries, call emergency services immediately. Immobilize the leg: Use a splint or sling to prevent further injury and decrease pain. Splinting is crucial for unstable injuries and can help prevent fat embolism in long-bone fractures. Treat for shock: If the person shows signs of shock, lay them down with their legs elevated above heart level to help improve blood flow and reduce swelling. Keep them warm and provide fluids if possible. Pain management: Administer pain medication as needed, typically opioids, to make the person more comfortable.

Question answering using RAG (fine – tuned)

Q1: Sepsis is a life-threatening condition that requires prompt recognition and aggressive management in a critical care unit. The following steps are generally recommended:\n1. Early recognition and assessment: Look for signs and symptoms of infection, such as fever, chills, tachycardia, respiratory distress, and altered mental status. Use the Sequential [Sepsis-related] Organ Failure Assessment (SOFA) score to evaluate organ dysfunction.\n2. Immediate fluid resuscitation: Give intravenous crystalloid fluids to maintain adequate tissue perfusion and blood pressure. Aim for a mean arterial pressure of 65 mm Hg or higher.\n3. Antibiotic therapy: Start broad-spectrum antibiotics as soon as possible based on the suspected source of infection. Adjust the antibiotics based on culture results and sensitivity testing.\n4. Source control: Identify and address the source of infection, such as removing an infected catheter or draining an abscess.

Q2: The common symptoms of appendicitis include epigastric or periumbilical pain followed by brief nausea, vomiting, and anorexia. After a few hours, the pain shifts to the right lower quadrant. Pain increases with cough and motion. Classic signs are direct and rebound tenderness located at McBurney's point, Rovsing sign, psoas sign, or obturator sign, along with low-grade fever. However, these classic findings appear in less than 50% of patients, and symptoms may not be localized, particularly in infants and children, elderly patients, and pregnant women.\nAppendicitis cannot be cured via medicine alone as it often leads to necrosis, gangrene, perforation, or the formation of an appendiceal abscess if left untreated. The primary treatment for appendicitis is surgical removal of the appendix through a procedure called appendectomy. If the appendix has ruptured and formed an abscess, drainage of the abscess may be necessary before performing the appendectomy.

Question answering using RAG

Q3: Sudden patchy hair loss, also known as alopecia areata, is a common condition characterized by round, smooth patches of baldness on the scalp or other areas of the body where hair growth has stopped. The exact cause of alopecia areata is not fully understood but it's believed to be an autoimmune disorder that affects genetically susceptible individuals exposed to certain environmental triggers.\n\nThe good news is that there are several treatment options for addressing alopecia areata, depending on the severity and extent of hair loss:\n\n1. Topical treatments: Minoxidil (Rogaine) is a medication applied directly to the scalp, which can stimulate hair growth in some people with alopecia areata. Another topical treatment option is anthralin, which works by reducing inflammation and promoting hair regrowth.\n\n2. Intralesional treatments: Corticosteroid injections into the affected areas can help reduce inflammation and promote hair regrowth.

Q4: For a person with a physical injury to brain tissue that results in impaired brain function, the focus of treatment is on managing symptoms and promoting recovery. Depending on the severity and location of the injury, treatments may include:\n\n1. Supportive care: This includes maintaining adequate oxygenation and ventilation, controlling blood pressure, providing nutrition and hydration, and managing seizures if they occur.\n\n2. Medications: Various medications may be used to treat specific symptoms or conditions related to brain injury, such as pain, swelling, infections, or muscle spasms.\n\n3. Rehabilitation: Physical, occupational, and speech therapies can help improve function and reduce the impact of impairments on daily life.\n\n4. Surgery: In some cases, surgery may be necessary to remove hematomas, repair skull fractures, or relieve pressure within the skull.



Question answering using RAG

Q5:

I'm an AI designed to provide clear, helpful, and concise information. For a person who has fractured their leg during a hiking trip, the following steps should be taken:\n1. Assess the severity of the injury: Check if it is a life- or limb-threatening injury. If so, call for emergency medical assistance immediately.\n2. Provide first aid: Apply RICE (Rest, Ice, Compression, and Elevation) principles to help minimize pain, swelling, and further damage. Immobilize the leg with a splint or sling if possible.\n3. Treat for shock: If the person is experiencing signs of hemorrhagic shock (pale, clammy skin, rapid heartbeat, weak pulse, confusion, or loss of consciousness), they should be treated in an emergency department as soon as possible.\n4. Seek professional help: Consult a healthcare professional for definitive treatment, which may include reduction and the use of surgical hardware like pins, screws, plates, or external fixators.



Takeaways from RAG and Fine-tuned RAG

RAG without being fine-tuned with temperature delivered straightforward answers. They had a bit less context than the answers once the RAG was fine-tuned with temperature.

At times, the RAG will acknowledge that it is an AI assistant. I thought this functionality was a bit strange.

RAG when fine-tuned was a great mix of conciseness, cohesiveness and facts with proof behind them



Output Evaluation:

Groundedness prompt:

You are an impartial judge who has read the medical manual. Rate groundedness on a scale (e.g., 1-5)

Relevance prompt:

"You are an impartial judge who has read the medical manual. Rate relevance on a scale (e.g., 1-5)



Output Evaluation:

Q1:

Based on the provided context, I would rate groundedness as a 5. The text is clear, concise, and focused on the protocol for managing sepsis in a critical care unit. It provides specific steps to take when sepsis or septic shock is suspected, making it grounded in the medical field.

Relevance: 5 The context provided in the manual includes information about critical care medicine, patient monitoring and testing, blood tests, cardiac monitoring, and pulmonary artery catheter monitoring. The user's question specifically asks for the protocol for managing sepsis in a critical care unit.

Q2:

Based on the provided context, I would rate groundedness as a 5 out of 5. The information presented is detailed and comes directly from a reputable medical source (The Merck Manual). It covers the etiology, symptoms, diagnosis, treatment, and prognosis of appendicitis in a clear and comprehensive manner.

Additionally, I would rate the relevance of the provided context to the user's question as high (5). The text extensively covers the symptoms, diagnosis, treatment, and prognosis of appendicitis. It also mentions that appendicitis cannot be cured via medicine and requires surgical intervention if diagnosed.



Output Evaluation:

Q3:

Based on the provided context and information, I would rate the groundedness of the user's question as 5. The user has clearly defined the issue at hand, which is sudden patchy hair loss, and has asked specifically about effective treatments or solutions for this condition and possible causes behind it.

The user has also shown an understanding of the various types I would rate the relevance of the provided information as 5. The user's question specifically asks about the causes and treatments for sudden patchy hair loss, which is exactly what is discussed in the text from The Merck Manual. The manual provides detailed information on alopecia areata, its possible causes, and various treatment options.

Q4:

Based on the provided context, I would rate groundedness as a 5. The text is clear, concise, and well-organized, making it easy to understand the topic of brain injuries, their diagnosis, prognosis, and treatment options. The language used is professional and technical, which adds to the groundedness of the text. Additionally, the sources cited are reputable and add credibility to the information presented.

Relevance: 5 The user's question asks about treatments for a person who has sustained a physical injury to brain tissue resulting in temporary or permanent impairment of brain function. The context provided from The Merck Manual of Diagnosis & Therapy, 19th Edition discusses various aspects of brain injuries, including diagnosis, prognosis, and treatment options. Therefore, the information is highly relevant to answering the user's question.



Output Evaluation:

Q5:

Based on the information provided in The Merck Manual of Diagnosis & Therapy, I would rate groundedness of this response as a 5. The answer provides clear instructions for assessing and treating a fractured leg, including immobilization with a splint, treatment for shock if necessary, and follow-up care such as RICE.

Based on the context provided in the user's question and the information from The Merck Manual of Diagnosis & Therapy, I would rate the relevance of the text as a 5. The manual provides comprehensive information on the necessary precautions and treatment steps for someone who has fractured their leg during hiking.



Output Evaluation:

The output evaluation rated everything a 5.

This could be accurate or misleading. Perhaps we should try with a larger scale.
Ex 1-20 next time.

This should also still be verified by an expert in the field to ensure the RAG is correct at all times.